TASK 21

- Migrate Database from On-premises to Cloud.

  -step 1: create EC2 instance

  -Step 2: Connect to CLI, install maridb105*, Start and Enable that service.

  -Step 3: Login to the database by Using Command → mysql -h localhost -u root -p (password).

  -Step 4: To assigned password Used Command → mysql_secure_installation

```
root@ip-172-31-40-215:~
[root@ip-172-31-40-215 ~]#
[root@ip-172-31-40-215 ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] n
 ... skipping.

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] n
 ... skipping.

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them.  This is intended only for testing, and to make the installation
go a bit smoother.  You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
 ... Success!

Normally, root should only be allowed to connect from 'localhost'.  This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] y
 ... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access.  This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
 - Dropping test database...
 ... Success!
 - Removing privileges on test database...
 ... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
 ... Success!

Cleaning up...
```

  -Step5: After that login to the Database Using Command → mysql -h localhost -u root -ppasssword

```
root@ip-172-31-40-215:~
[root@ip-172-31-40-215 ~]# mysql -h localhost -u root -pachal12345
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 9
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

  -Step 6: show databases; →used command for show database.

```
[root@ip-172-31-40-215 ~]# mysql -h localhost -u root -pachal12345
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 9
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
3 rows in set (0.000 sec)
```

-Step 7: create database fct; → used command to create database.

-Step 8: use database fct → command for used database.

-Step 9: create table student_info(rollno int, name varchar(20), marks float);

```
root@ip-172-31-40-215:~
MariaDB [(none)]> use fct;
Database changed
MariaDB [fct]> show table;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near '' at line 1
MariaDB [fct]> show tablse;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'tablse' at line 1
MariaDB [fct]> show tables;
Empty set (0.000 sec)

MariaDB [fct]> create table student_info(rollno int, name varchar(20), marks float);
Query OK, 0 rows affected (0.008 sec)

MariaDB [fct]> show table;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near '' at line 1
MariaDB [fct]> show tables;
+--------------+
| Tables_in_fct |
+--------------+
| student_info  |
+--------------+
1 row in set (0.000 sec)
```

-Step 10: insert into student_info values(Give value here); → command for insert value into table.

```
MariaDB [fct]> insert into student_info values(1, "Achal", 90),(2, "Nikita", 92),(3, "Divya", 93),(4, "Mansi", 90),(5, "Dhanno", 87),(6, "Divyani", 87);
Query OK, 6 rows affected (0.002 sec)
Records: 6  Duplicates: 0  Warnings: 0

MariaDB [fct]> show tables;
+---------------+
| Tables_in_fct |
+---------------+
| student_info  |
+---------------+
1 row in set (0.000 sec)

MariaDB [fct]> select * from student_info
    -> ;
+--------+---------+-------+
| rollno | name    | marks |
+--------+---------+-------+
|      1 | Achal   |    90 |
|      2 | Nikita  |    92 |
|      3 | Divya   |    93 |
|      4 | Mansi   |    90 |
|      5 | Dhanno  |    87 |
|      6 | Divyani |    87 |
+--------+---------+-------+
6 rows in set (0.000 sec)

MariaDB [fct]> |
```

-Step 11: Show table values used command for that → select * from student_info;

-Step 12: Take Backup of that database using command →

mysqldump -u username -ppassward databasename > filename

```
root@ip-172-31-40-215:~
[root@ip-172-31-40-215 ~]# mysqldump -u root -p fct > student.sql
Enter password:
[root@ip-172-31-40-215 ~]# ls
student.sql
[root@ip-172-31-40-215 ~]#
```

```
root@ip-172-31-40-215:~
[root@ip-172-31-40-215 ~]# mysqldump -u root -p fct > student.sql
Enter password:
[root@ip-172-31-40-215 ~]# ls
student.sql
[root@ip-172-31-40-215 ~]# cat student.sql
/*M!999999\- enable the sandbox mode */
-- MariaDB dump 10.19  Distrib 10.5.29-MariaDB, for Linux (x86_64)
--
-- Host: localhost    Database: fct
-- ------------------------------------------------------
-- Server version       10.5.29-MariaDB

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8mb4 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;

--
-- Table structure for table `student_info`
--

DROP TABLE IF EXISTS `student_info`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8mb4 */;
CREATE TABLE `student_info` (
  `rollno` int(11) DEFAULT NULL,
  `name` varchar(20) DEFAULT NULL,
  `marks` float DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=latin1 COLLATE=latin1_swedish_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `student_info`
--

LOCK TABLES `student_info` WRITE;
/*!40000 ALTER TABLE `student_info` DISABLE KEYS */;
INSERT INTO `student_info` VALUES (1,'Achal',90),(2,'Nikita',92),(3,'Divya',93),(4,'Mansi',90),(5,'Dhanno',87),(6,'Divyani',87);
/*!40000 ALTER TABLE `student_info` ENABLE KEYS */;
UNLOCK TABLES;
/*!40103 SET TIME_ZONE=@OLD_TIME_ZONE */;

/*!40101 SET SQL_MODE=@OLD_SQL_MODE */;
/*!40014 SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS */;
/*!40014 SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS */;
/*!40101 SET CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
/*!40101 SET CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
/*!40101 SET COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;
/*!40111 SET SQL_NOTES=@OLD_SQL_NOTES */;

-- Dump completed on 2025-12-15 10:41:04
[root@ip-172-31-40-215 ~]#
```

Step 13: login RDS instance to this server create one database.

```
root@ip-172-31-40-215:~
[root@ip-172-31-40-215 ~]# mysql -h database-1.cn68022gealf.us-east-2.rds.amazonaws.com -u root -pachal12345
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 45
Server version: 11.4.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| innodb             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.005 sec)

MariaDB [(none)]> create database fct;
Query OK, 1 row affected (0.005 sec)

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| fct                |
| information_schema |
| innodb             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.001 sec)

MariaDB [(none)]>
```

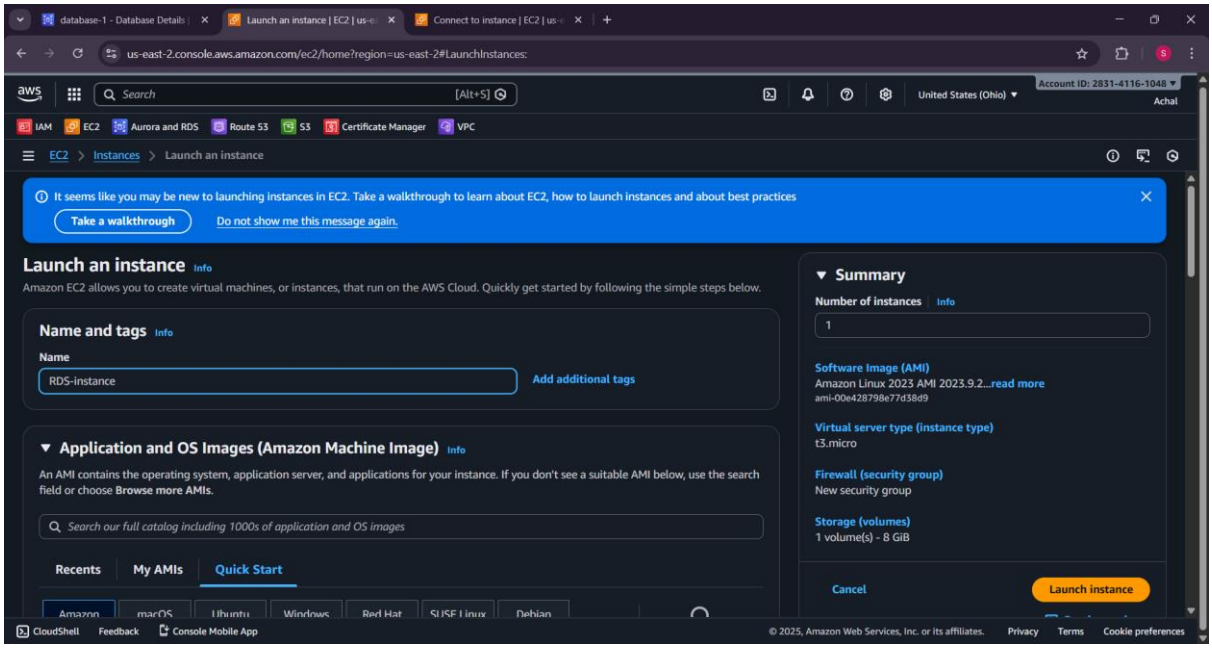
-Step 14: Now Migrate that file to the RDS instance using command. →

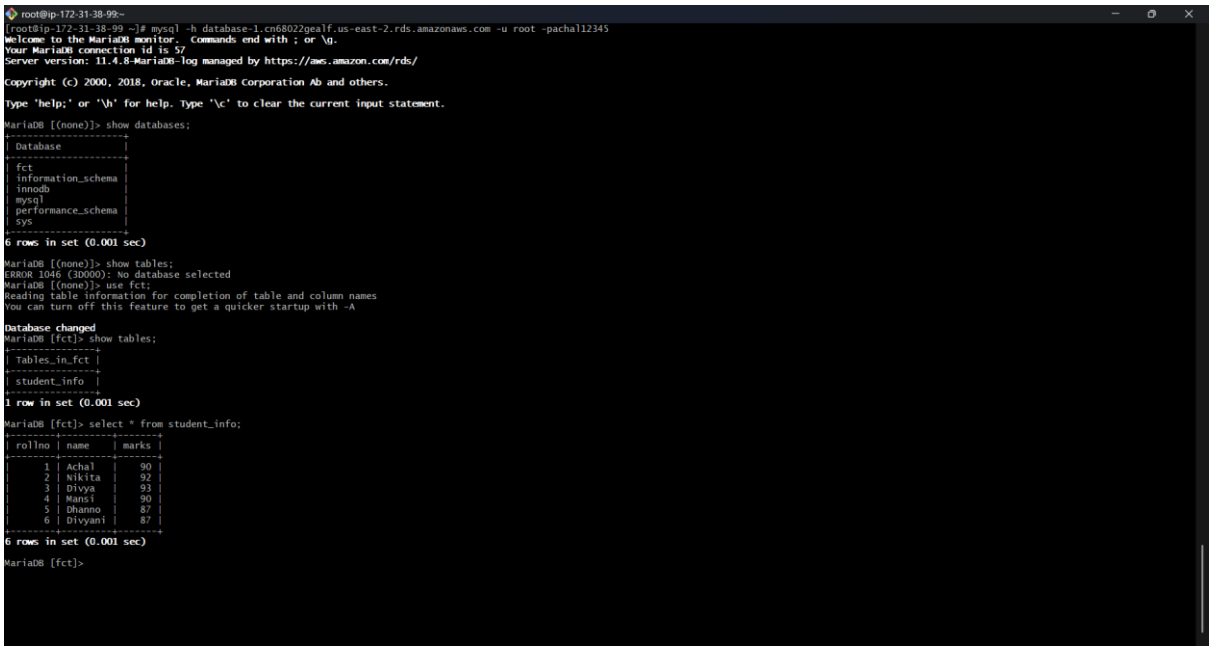
mysql -h endpoint -u username -p databasename < filename

```
root@ip-172-31-40-215:~
[root@ip-172-31-40-215 ~]# mysql -h database-1.cn68022gealf.us-east-2.rds.amazonaws.com -u root -pachal12345
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 45
Server version: 11.4.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| innodb             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.005 sec)

MariaDB [(none)]> create database fct;
Query OK, 1 row affected (0.005 sec)

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| fct                |
| information_schema |
| innodb             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.001 sec)

MariaDB [(none)]> ^DBye
[root@ip-172-31-40-215 ~]# mysql -h database-1.cn68022gealf.us-east-2.rds.amazonaws.com -u root -p fct < student.sql
Enter password:
[root@ip-172-31-40-215 ~]# |
```

-Step 15: Logout that database instance.

-Step 16: Create another ec2-instance and check all values are migrate or not to the RDS instance all values will migrate to the RDS instance.


database-1 - Database Details
Launch an instance | EC2 | us-e
Connect to instance | EC2 | us-e
us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#LaunchInstances:
aws
Search
[Alt+S]
United States (Ohio)
Account ID: 2831-4116-1048
Achal
IAM
EC2
Aurora and RDS
Route 53
S3
Certificate Manager
VPC
EC2 > Instances > Launch an instance
It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices
Take a walkthrough
Do not show me this message again.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
RDS-instance
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
My AMIs
Quick Start
Amazon
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-00e428798e77d38d9
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


root@ip-172-31-38-99:~
[root@ip-172-31-38-99 ~]# mysql -h database-1.cn68022gealf.us-east-2.rds.amazonaws.com -u root -pachal12345
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 57
Server version: 11.4.8-MariaDB-log managed by https://aws.amazon.com/rds/
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
MariaDB [(none)]> show databases;
+--------------------+
| Database |
+--------------------+
| fct |
| information_schema |
| innodb |
| mysql |
| performance_schema |
| sys |
+--------------------+
6 rows in set (0.001 sec)
MariaDB [(none)]> show tables;
ERROR 1046 (3D000): No database selected
MariaDB [(none)]> use fct;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
Database changed
MariaDB [fct]> show tables;
+---------------+
| Tables_in_fct |
+---------------+
| student_info |
+---------------+
1 row in set (0.001 sec)
MariaDB [fct]> select * from student_info;
+--------+---------+-------+
| rollno | name | marks |
+--------+---------+-------+
1	Achal	90
2	Nikita	92
3	Divya	93
4	Mansi	90
5	Dhanno	87
6	Divyani	87
+--------+---------+-------+
6 rows in set (0.001 sec)
MariaDB [fct]>